

Unit 1: Equivalent Algebraic Expressions

Learning Goals:

- To simplify rational expressions by factoring, and state the restrictions on the variable values;
- To add, subtract, multiply, and divide rational expressions;
- To determine if two given algebraic expressions are equivalent.

Self-Assessment

	H	R	G
To simplify rational expressions by factoring, and state the restrictions on the variable values;			
To add, subtract, multiply, and divide rational expressions;			
To determine if two given algebraic expressions are equivalent.			

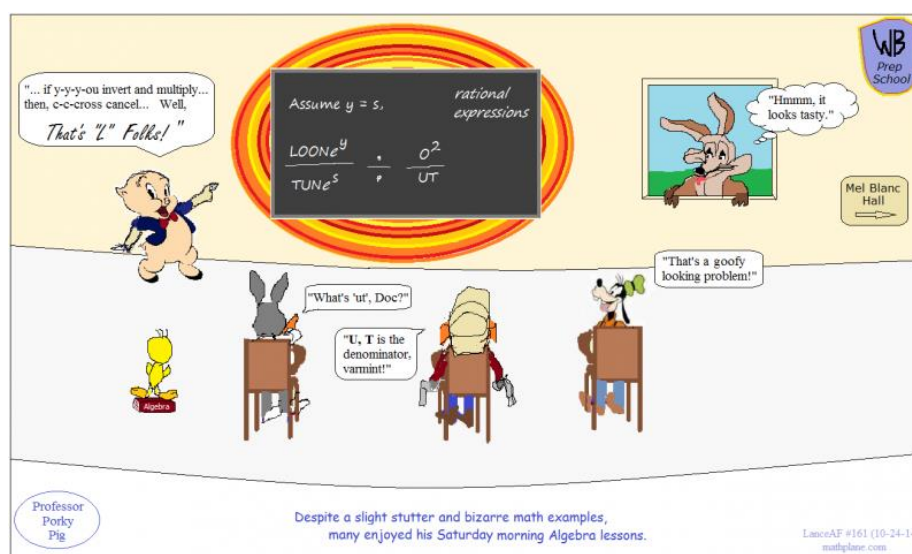
H – I need to ask for **HELP** with this.

R – I need to **REVIEW** this.

G – I've **GOT THIS** down.

Homework Log:

Lesson	Section	Assigned Work	Extra Practice
1	2.1 & 2.2 Adding, Subtracting & Multiplying Polynomials	Text p. 88 #6, 7, 10, 11 Text p. 95 #7 – 13	Text p. 88 #4, 5 Text p. 95 #4, 5, 6
2	2.3 Factoring Polynomials	Text p. 102 #1 – 7, 9, 14ab	Worksheet online
3	2.4 Simplifying Rational Functions	Text. p. 112 #1 – 5, 8 – 14	Worksheet online
4	2.4 Continued	LP: pgs. 11 – 12 (eoo for each)	
5	2.6 Multiplying & Dividing Rational Expressions	Text. p. 121 #3 – 10	Worksheet online
6	2.7 Adding & Subtracting Rational Expressions	Text. p. 128 #1 – 10 (eoo for each)	Worksheet online
7	Review for Test	LP. pp. 17 – 18	Text. p. 132 #4 – 10, 12 – 15



2.1 & 2.2 Adding, Subtracting and Multiplying Polynomials

Definition:

Like Terms	Terms that contain the same variables taken to the same powers.
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Which of the following are like terms?

$5x^2$

$2x^2y$

$-8x^2$

$3xy$

$12x^2$

$9x^2y$

Add or subtract	To add or subtract polynomials, combine like terms.
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Method 1: HORIZONTALLY



A $(5x + 14) + (2x^2 - 4x + 3)$

STEPS
1. Eliminate the brackets.
2. Combine like terms.
3. Write in standard form.

B $(x^2 + 4x) - (2x^2 - 5x + 3)$

STEPS
1. Distribute the negative into the second the bracket.
2. Follow same steps as addition.

Method 2: VERTICALLY



C $(-4x^3 + 2x^2 + 9) + (7x^3 + 10x^2 - 3x - 4)$

STEPS
1. Write one polynomial under the other so that like terms appear in the same column.
2. Add like terms.

D $(3x^4 - 10x^2 - 6x + 1) - (2x^4 + 5x^3 + 6x + 1)$

STEPS
1. Write one polynomial under the other so that like terms appear in the same column.
2. Subtract like terms.

Multiply	To multiply polynomials, use the distributive property. • Multiply the coefficients. • Add the exponents.
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A Monomial x Monomial

$$(-5xy^3)(8y^4z^3)$$

B Monomial x Polynomial

$$4x(3x^2 - 8x - 4)$$

C Binomial x Binomial

$$(3x + 2)(x - 1)$$

D Binomial x Polynomial

$$(x + 2)(x^2 - 3x - 6)$$

Practice: Simplify.

1. $(x^3 + 3x^2 + 2) + (2x^3 + 4)$

2. $(x^4 + 3x^2 - 6x + 2) - (15x^4 + 2x^3 - 6x^2 + 5x - 3)$

3. $(2x)(7y)(5x - 3y)$

4. $-2(5x + 1)(x^2 - 3x + 8)$

5. $4(5x + 2) - 3(2x - 7)$

6. $(x - 9)^2 + (x + 4)(x + 6)(x - 1)$

Application Example:

A rectangle is twice as long as it is wide. Write a simplified expression for the change in area if the length is increased by 1 unit and the width is decreased by 1 unit.

2.3 Factoring Polynomials

Factoring a polynomial involves breaking the polynomial down into a product of simpler polynomials.

Common Factoring: Divide out the greatest common factor between the terms.

a) $16m^2n^4 + 8m^2n^3 - 4m^2n^2$

b) $3(x + 4) + (3x + 5)(x + 4)$

Sum and Product: $ax^2 + bx + c$, $a = 1$

Determine two numbers that add to b and multiply to c . Note: Always look for GCF first!

a) $x^2 + 6x + 8$

b) $2x^2 - 4x - 16$

Criss-Cross: $ax^2 + bx + c$, $a \neq 1$

Determine the numbers that multiply to a and two numbers that multiply to c . Cross multiply the numbers and add to get b .

a) $3x^2 - 14x - 5$

b) $5a^3b - 12a^2b + 4ab$

Perfect Square Trinomials:

Trinomials that are in the form, $A^2 \pm 2AB + B^2$, can be factored as $(A \pm B)^2$.

a) $x^2 - 10x + 25$

a) $16x^2 + 24x + 9$

Difference of Squares:

When one squared term is subtracting another squared term, square root each term. Add the terms in one bracket and subtract them in the other bracket.

a) $81x^2 - 49y^2$

b) $(x - y)^2 - 16$

Factoring by Grouping:

Group terms that have a common factor. Factor each grouping to produce a new common factor that is then factored out.

a) $ax - ay + bx - by$

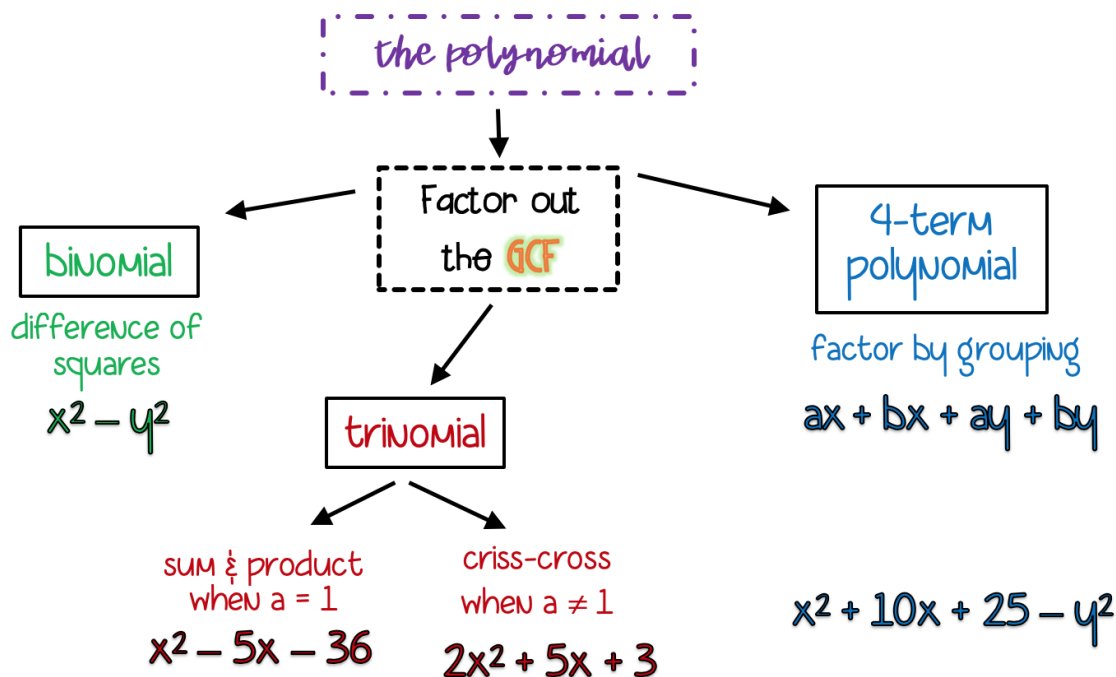
b) $2ax + 3by + 3bx + 2ay$

Group terms that make a perfect square trinomial. Factoring the perfect square trinomial should then produce a difference of square which can then be factored.

c) $1 - 16y^2 + 6x + 9x^2$

d) $a^2 - b^2 + 8bc - 16c^2$

Factoring Flow Chart



Memorize the Squares

$1^2 =$	$2^2 =$	$3^2 =$	$4^2 =$	$5^2 =$
$6^2 =$	$7^2 =$	$8^2 =$	$9^2 =$	$10^2 =$
$11^2 =$	$12^2 =$	$13^2 =$	$14^2 =$	$15^2 =$
$16^2 =$	$17^2 =$	$18^2 =$	$19^2 =$	$20^2 =$
$21^2 =$	$22^2 =$	$23^2 =$	$24^2 =$	$25^2 =$
$26^2 =$	$27^2 =$	$28^2 =$	$29^2 =$	$30^2 =$

2.4 Simplifying Rational Expressions

Goal: to simplify a rational expression and state restrictions on the denominator

Rational Expressions

If x and y are two integers, then the quotient $\frac{x}{y}$ is a **rational number** with the restriction $y \neq 0$.

Examples of rational numbers: a) $\frac{3}{5}$ b) $-8 = \frac{-8}{1}$ c) $0.13 = \frac{13}{100}$ d) $0.\overline{7} = \frac{7}{9}$

If P and Q are polynomials, then the quotient $\frac{P}{Q}$ is a **rational expression** with the restriction $Q \neq 0$.

Examples of rational expressions: a) $\frac{1}{xy}$ b) $\frac{3+x}{5-x}$ c) $\frac{x+1}{3x-2}$ d) $\frac{-2x^2}{x^2+4x-5}$

Restrictions:

Much like rational numbers, rational expressions cannot have denominators that equal 0 since **division by 0 is undefined**. We must, therefore, place restrictions on the variables in the denominators of rational expressions.

Example 1: State the restriction for each rational expression.

a) $\frac{5}{x}$ b) $\frac{x}{x+4}$ c) $\frac{x+1}{3x-2}$ d) $\frac{5x}{(x-7)(x+2)}$

In some cases, the denominator is not in factored form making it difficult to determine the restrictions. Therefore, the **denominator must be factored first** before stating restrictions.

e) $\frac{x+5}{x^2-2x}$ f) $\frac{3x}{9-x^2}$ g) $\frac{x^2-4}{x^2-7x-8}$

Simplifying Rational Expressions

To simplify rational expressions, we use the same techniques as with simplifying rational numbers.

Recall: Simplify $\frac{30}{42}$

$$= \frac{(2)(3)(5)}{(2)(3)(7)}$$

$$= \frac{5}{7}$$

- factor the numerator and denominator into prime factors.
- divide both numerator and denominator by factors common to both.

To simplify rational expressions: 1) **factor** both numerator and denominator

2) **state restrictions** on any variables in the denominator

3) divide by any **common factors**

Example 2: Simplify the following rational expressions. State any restrictions.

a) $\frac{4x^4 - 72x^3 + 16x^2}{4x^2}$

b) $\frac{36x^2y}{-16xy^2}$

c) $\frac{m^2 - 4m}{3m^2 - 12m}$

d) $\frac{12 - 4x}{2x - 6}$

e) $\frac{x^2 - 2x - 15}{x^2 - x - 12}$

f) $\frac{2a^2 + 5a - 3}{2a^2 - 7a + 3}$

Application of Rational Expressions

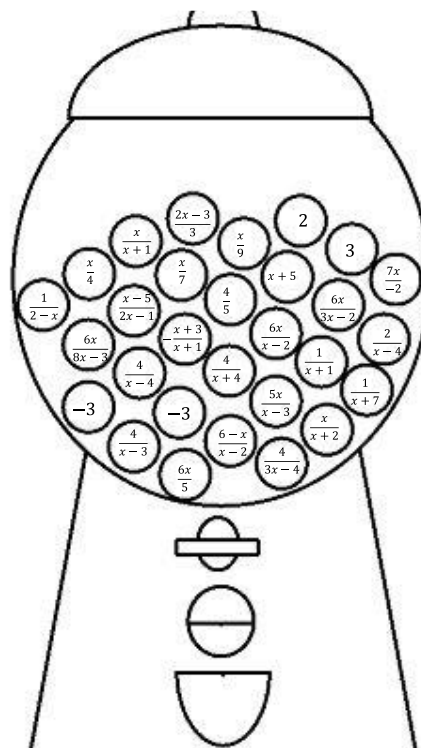
The concentration (mg) of pain-killing drug A in the bloodstream, t hours after it is taken orally, is given by $C_A = \frac{5t^4 - 10t^2}{t^5 - 4t}$ and the concentration (mg) of pain-killing drug B in the blood stream, t hours after it is taken is given by the expression $C_B = \frac{3t^2 - 12t}{(t-4)(2t^2+5)}$.

- a) Simplify both expressions completely.
- b) After 6 hours, which drug has the lowest concentration in the bloodstream?

Simplify Rational Expression Colour Me Gumball

Simplify and color the matching answer. If any portion of the picture is not numbered then it is your choice how to color.

1. RED $\frac{3x - 27}{x - 9}$	2. BLUE $\frac{x + 1}{x^2 + 2x + 1}$	3. YELLOW $\frac{x^2 - 4x}{7(x - 4)}$	4. GREEN $\frac{5 - x}{x^2 - 7x + 10}$
5. RED $\frac{2x^2 - x - 3}{3x + 3}$	6. BLUE $\frac{x^2 - x - 30}{x - 6}$	7. YELLOW $\frac{x^2 - 2x - 15}{2x^2 + 5x - 3}$	8. GREEN $\frac{x^2 - 3x}{x^2 - 2x - 3}$
9. RED $\frac{4x + 16}{x^2 - 16}$	10. BLUE $\frac{x - 7}{x^2 - 49}$	11. YELLOW $\frac{9 - x^2}{x^2 - 2x - 3}$	12. GREEN $\frac{6x^2 - 12x}{x^2 - 4x + 4}$



Name _____ Date _____ Period _____

Simplifying Rational Algebraic Expressions: Maze

Instructions: Simplify each rational expression. Follow the path of correct answers to complete the maze.

Begin:

$\frac{9x}{27x}$	$\frac{x}{3}$	$\frac{16x^3}{44x^6}$	$\frac{4x}{11}$	$\frac{x-2}{2x-4}$	$\frac{1}{2}$	$\frac{12x^4}{32x^2}$	$\frac{4x^2}{8}$	$\frac{27}{81x}$
$\frac{1}{3}$	$\frac{2x}{3}$	$\frac{4x^2}{11x^4}$	$\frac{7}{13}$	$\frac{14x}{26x}$	$\frac{2}{3x}$	$\frac{3x^2}{8}$	3	$\frac{1}{6}$
$\frac{18x}{24}$	$\frac{2x}{3}$	$\frac{24}{15x-3}$	$\frac{8}{5x-1}$	$\frac{7x}{13}$	$\frac{3+6x}{11}$	$\frac{3x+12}{x+4}$	$\frac{27x+18}{27}$	$\frac{3x+2}{3}$
$\frac{3x}{4}$	$\frac{9}{3x+2}$	$\frac{2x+3}{4}$	$\frac{3x}{4}$	$\frac{-2x+6}{4x-14}$	$\frac{-2}{x+7}$	$\frac{9+66x}{21x+12}$	$\frac{2x^2}{7}$	$\frac{14x^4}{42x^2}$
$\frac{45}{15x+10}$	$\frac{x}{3+6x}$	$\frac{2x+1}{3}$	$\frac{4x+16}{24x+8}$	$\frac{x+3}{4}$	$\frac{6x-1}{3x}$	$\frac{-12x+2}{-6x}$	$\frac{3x}{2}$	$\frac{x^2}{3}$
$\frac{7x}{21+42x}$	$\frac{7x}{3+6x}$	$\frac{4x+16}{24x+8}$	$\frac{11}{4}$	$\frac{30x+12}{62+34x}$	$\frac{3x+7}{30x+5}$	$\frac{15x+40}{60x+15}$	$\frac{3}{8x+25}$	$\frac{12x+9}{18x+12}$
$\frac{2}{3}$	$\frac{x+4}{6x+2}$	$\frac{40x+24}{56x+72}$	$\frac{4x+1}{3}$	$\frac{15x+6}{17x+31}$	$\frac{5x+4}{9x+3}$	$\frac{20x+25}{40+35x}$	$\frac{4x+5}{8+7x}$	$\frac{4x+3}{6x+4}$
$\frac{20x}{35x^2}$	$\frac{5x+3}{7x+9}$	$\frac{6}{7}$	$\frac{2x}{x+4}$	$\frac{x+7}{5x+35}$	$\frac{1}{5}$	$\frac{3+10x}{31+12x}$	$\frac{9+10x}{31+2x}$	$\frac{12}{32x+100}$
$\frac{4}{7x}$	$\frac{x^2+3}{6x^2+18}$	$\frac{1}{6}$	$\frac{3x^3+6x^2}{3x^3+27x^2}$	$\frac{26x+13}{4+8x}$	$\frac{1}{4}$	$\frac{9+30x}{93+36x}$	$\frac{6}{3x+10}$	$\frac{18+2x}{36+4x}$
$\frac{49x}{7x+28}$	$\frac{7x}{x+4}$	$\frac{3x^3+6x^2}{3x^3+27x^2}$	$\frac{x+2}{x+9}$	$\frac{26x+13}{4+8x}$	$\frac{1}{4}$	$\frac{9+30x}{93+36x}$	$\frac{9+10x}{31+2x}$	$\frac{1}{2}$
END								

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Simplifying Rational Expressions Practise

In each of the following, state any restrictions on the variables.

1. Simplify

a) $\frac{3t^3 + 6t^2 - 15t}{3t}$

b) $\frac{6a^2 + 9a}{12a^2}$

d) $\frac{14n^4 - 4n^3 + 6n^2 + 8n}{2n^2}$

e) $\frac{4m^2 - 8mn}{4mn}$

g) $\frac{16a^2bc}{14a^2b^2c^2}$

h) $\frac{-4x^4y^2z}{20x^3y^3z}$

2. Express in simplest form.

a) $\frac{5x}{5(x+4)}$

b) $\frac{8t^2(t+5)}{4t(t-5)}$

d) $\frac{(m-1)(m+2)}{(m+4)(m-1)}$

e) $\frac{2x}{2x+8}$

g) $\frac{10x}{5x^2 - 15x}$

h) $\frac{4x}{16x^3 - 12x}$

3. Simplify.

a) $\frac{6t-36}{t-6}$

b) $\frac{4m+24}{8m-24}$

d) $\frac{a^2+2a}{a^2-3a}$

e) $\frac{8x^2+4x}{6x^2+3x}$

g) $\frac{4x+4y}{5x+5y}$

h) $\frac{4a^2b+8ab}{6a^2-6a}$

4. Express in simplest equivalent form.

a) $\frac{m-2}{m^2-5m+6}$

b) $\frac{y^2+10y+25}{y+5}$

d) $\frac{x^2-4}{5x+10}$

e) $\frac{a^2+a}{a^2+2a+1}$

g) $\frac{2w+2}{2w^2+3w+1}$

h) $\frac{3t^2-8t+4}{6t^2-4t}$

j) $\frac{5x^2+3xy-2y^2}{3x^2+3xy}$

5. Simplify.

a) $\frac{y-2}{2-y}$

b) $\frac{3-x}{x-3}$

d) $\frac{6-10w}{15w-9}$

e) $\frac{x^2-1}{1-x^2}$

6. Simplify.

a) $\frac{x^2+4x+4}{x^3+5x+6}$

b) $\frac{x^2-a-12}{a^2-9a+20}$

d) $\frac{y^2-25}{y^2-8y+15}$

e) $\frac{x^2-12x+36}{x^2-n-6}$

g) $\frac{p^2-16}{p^2-3t+2}$

h) $\frac{2t^2-t-1}{t^2-3t+2}$

j) $\frac{6x^2-13x+6}{8x^2-6x-9}$

k) $\frac{3z^2-7z+2}{9z^2-6z+1}$

7. **Saskatchewan flag** The area of a Saskatchewan flag can be represented by the polynomial $x^2 + 3x + 2$ and its width by $x + 1$.

a) Write a rational expression that represents the length.

b) Write the expression in simplest form.

c) If x represents 1 unit of length, what is the ratio length:width for a Saskatchewan flag?

8. Simplify, if possible.

a) $\frac{1-x}{x-1}$

b) $\frac{x-1}{x+1}$

d) $\frac{3t-7}{3t-7}$

e) $\frac{t^2-s^2}{(s+t)^2}$

a) $\frac{2x-y}{x-y}$

b) $\frac{4x}{3x+y}$

d) $\frac{x^2-8}{x^3-8}$

e) $\frac{x^2+3x-11}{x^2-x-1}$

9. For which values of x are the following rational expressions not defined?

a) $\frac{2x-y}{x-y}$

b) $\frac{3x+y}{x^3}$

c) $\frac{3x^2+5xy+2y^2}{4x^2-9y^2}$

c) $\frac{2x+6}{x^2-6x-27}$

f) $\frac{x^2-9}{2x^2y-6xy}$

i) $\frac{8z+6z^2}{9z^2-16}$

c) $\frac{2t-1}{4-8t}$

Solutions

1. a) $\frac{t^2+2t-5}{2t^2+3t-4}$, $a \neq 0$, $t \neq 2$, $t \neq -3$, $t \neq 0$
 b) $\frac{3x}{x^2}$, $x \neq 0$
 c) $\frac{3x}{x^2}$, $a, b, c \neq 0$
 d) $\frac{3(m-4)}{2t(t+5)}$, $m \neq 0$, $x \neq -4$, $t \neq -5$
 e) $\frac{1}{2x}$, $x \neq 0$, $m \neq 1$, $-4 \neq x$
 f) $\frac{y+2}{x}$, $y \neq 0$, $3 \neq x$
 g) $\frac{2x-4y}{m}$, $x \neq 0$, $x \neq 2y$
 h) $\frac{1}{2x-4y}$, $x \neq 0$, $x \neq 2y$
 i) $\frac{3}{5}$, $\frac{3}{5} \neq 2$
 j) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 k) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
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 bc) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bd) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 be) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bf) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bg) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bh) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
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 bm) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bn) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bo) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bp) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bq) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 br) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bs) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bt) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bu) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bv) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bw) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bx) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 by) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 bz) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 ca) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 cb) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 cc) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 cd) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 ce) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 cf) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 cg) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 ch) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 ci) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
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 cz) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 da) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
 db) $\frac{5}{3}$, $\frac{5}{3} \neq 2$
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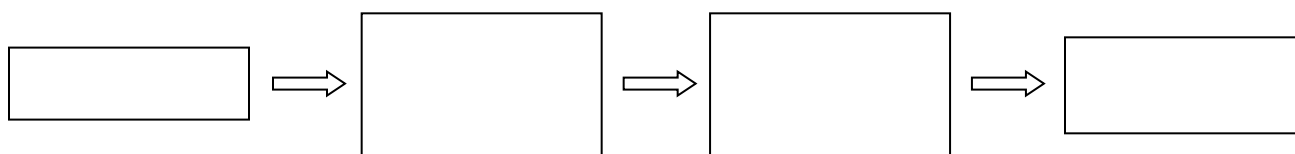
2.6 Multiplying and Dividing Rational Expressions

Rational expressions can be multiplied in the same way that fractions are multiplied.

Example: $\frac{3}{4} \cdot \frac{5}{6}$

Take Note:

When we multiply rational expressions, our steps are as follows:



Simplify each of the following. State the restrictions on the variables.

a) $\frac{9x^2}{4xy} \cdot \frac{12xy^2}{3x}$

b) $\frac{x^2 - 3x + 2}{x^2 + 3x + 2} \cdot \frac{8x + 8}{4x - 8}$

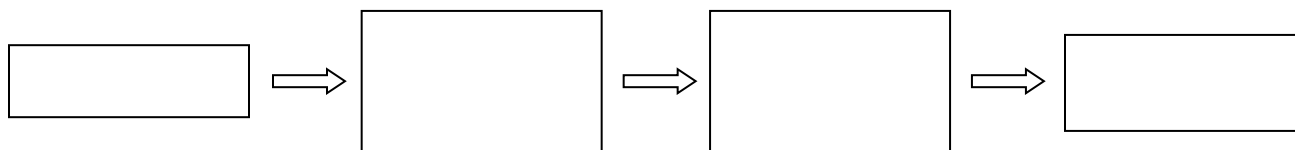
c) $\frac{2x^2 + 5x + 2}{4x^2 - 8x - 5} \cdot \frac{2x^2 - 11x + 15}{3x^2 + 7x + 2}$

Rational expressions can be divided in the same way that fractions are divided.

Example: $\frac{2}{3} \div \frac{5}{7}$

Take Note:

When we divide rational expressions, our steps are as follows:



Simplify each of the following. State the restrictions on the variables.

a) $\frac{8x^2 + 2x - 1}{6x^2 - x - 2} \div \frac{8x - 2}{9x - 6}$

b) $\frac{2x + 1}{1 - 3x} \div \frac{4x^2 - 1}{9x^2 - 1}$

2.7 Adding and Subtracting Rational Expressions

Rational expressions can be added or subtracted in the same way that fractions are added or subtracted.

Recall, $\frac{1}{5} + \frac{3}{4}$

Examples: Simplify each of the following. State any restrictions on the variables.

a) $\frac{3a}{4} + \frac{a}{2} - \frac{5a}{6}$

b) $\frac{2x - 1}{6} - \frac{x - 4}{8}$

Adding or Subtracting with Monomial Denominators

c) $\frac{5}{x^3} + \frac{3}{2x^2} - \frac{7}{3x}$

d) $\frac{3}{2m^2n} - \frac{1}{m^2n^3} + \frac{4}{5mn}$

Denominators with Binomial Factors

$$\text{e)} \quad \frac{2t}{t-5} - \frac{t}{3t-15}$$

$$\text{f)} \quad \frac{3}{x+1} + \frac{4}{x-2}$$

$$\text{g)} \quad \frac{2t}{t^2-4} + \frac{5}{t+2}$$

$$\text{h)} \quad \frac{x+1}{x^2+2x-3} + \frac{x+2}{x^2+4x-5}$$

$$\text{i)} \quad \frac{2x-1}{2x^2+3x+1} + \frac{2x+1}{3x^2+4x+1} \cdot \frac{3x+1}{x+2}$$

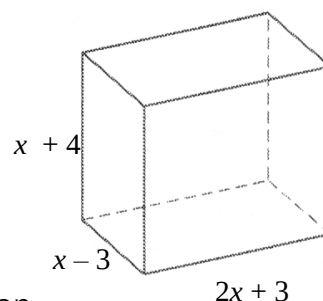
Relax... Review Unit 1!

1. The perimeter of a triangle is $5x - 2y + 3z$. If two sides have lengths $3y + z$ and $4x - y + z$, what is the length of the third side?

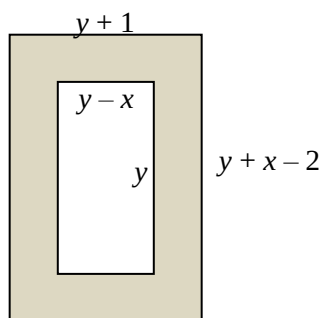
2. The dimensions of a rectangular prism are represented by binomials, as shown.

a) Write a simplified expression that represents the surface area of this prism.

b) Write a simplified expression that represents the volume of this prism.



3. Write and simplify an expression to represent the area of the shaded region.



4. The area of a Saskatchewan flag can be represented by the polynomial $x^2 + 3x + 2$ and its width by $x + 1$. Write a rational expression that represents the length, in simplest form.

5. An isosceles triangle has two sides of length $9x + 3$. The perimeter of the triangle is $30x + 10$.

a) Determine the ratio of the base to the perimeter, in simplified form. State the restriction on x .

b) Explain why the restriction on x in part (a) is necessary in this situation.

6. A rectangle is six times as long as it is wide. Determine the ratio of its area to its perimeter, in simplest form, if its width is w .

7. A rectangle has a width of $\frac{x^2 - 9}{x}$ and a length of $\frac{x^3 - 4x}{x^2 + 5x + 6}$. Write the area of the rectangle in simplest form.

8. Two triangles have the same base length, represented by x . The height of one triangle is $x + 1$. The height of the other triangle is $x + 3$. Write and simplify an expression that represents the total area of the two triangles.

9. The quotient of two polynomials is $3x - 2$. Give an example of a rational expression equivalent to this polynomial that has the restriction $x \neq 4$.

10. Give an example of a rational function that could have three restrictions that are consecutive numbers.
11. Is $(2x^2 + 3x + 1)^2 = 4x^4 + 9x^2 + 1$? Justify your answer by showing work (ie. Perform a LS/RS check.).
12. Fully factor each of the following.
- | | | |
|-----------------------------|----------------------------------|---------------------------|
| a) $3x^5 - 12x^3 - x^2 + 4$ | b) $x^2 - 8x + 16 - 4y^2$ | c) $(x+2)^2 - 3(x+2) + 2$ |
| d) $49m^2 - (x - y)^2$ | e) $fx - gy + gx - fy$ | f) $10x^2 + 38x + 20$ |
| g) $g(1 - x) - gx + gx^2$ | h) $36(2x - y)^2 - 25(x - 2y)^2$ | i) $x^4 - x^3 + x^2 - x$ |

Answers

1. $x - 4y + z$ 2. a) $10x^2 + 18x - 18$ b) $2x^3 + 5x^2 - 21x - 36$ 3. $2xy + x - y - 2$ 4. $x + 2; x^1 - 1$
5. a) $\frac{2}{5}$ 6. $\frac{3w}{7}$ 7. $x^2 - 5x + 6$ 8. $x^2 + 2x$ 12. a) $(3x^3 - 1)(x + 2)(x - 2)$ b) $(x - 4 + 2y)(x - 4 - 2y)$
- c) $x(x + 1)$ d) $(7m - x + y)(7m + x - y)$ e) $(f + g)(x - y)$ f) $2(5x^2 + 19x + 10)$ g) $g(x - 1)^2$
- h) $(17x - 16y)(7x + 4y)$ i) $x(x - 1)(x^2 + 1)$